Set Theory

Suppose that you roll a standard 6-sided die.

1. On a given roll, what values could you expect to see?

2. How often would you expect to see even numbers?

3. If another die was rolled,

The previous example illustrates several components of *Set Theory*, which forms the foundations of probability.

> Definition
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> The *sample space* of an experiment, S, is a collection of all possible outcomes of the experiment.

Using a single die toss, the set sample space is

$S =$

Now suppose that two identical 6-sided dice are tossed. The new sample space is given by

$S =$

Definition

An *event* can be defined as any subset of the sample space, including the entire sample space itself.

Events can be defined as any combination of elements within the sample space. For example, when rolling two dice, the events could be defined as $\{1, 2\}$ or when the sum of the dice is greater than 10.

Given any two or more events, there are several set operators that we need to know.

Definitions

Let A and B be two events within S.
If A is a **subset** of B, $A \subseteq B$, then all elements within A are also within B.
The **union** of A and B, $A \cup B$, is the set of elements belonging to either A or B.
The **intersection** of A and B, $A \cap B$, is the set of elements belonging to both A and B.
The **complement** of A, written as A^c, is the set of all elements that are not in A.

Sketch of operators

In-class Exercises

Probability Theory

Note

Given a sample space S, a probability function is a function P such that all of the following are satisfied:

1. $P(A) \geq 0$ for all events A in the sample space.
2. $P(S) = 1$
3. If events $A_1, A_2, ...$ are pairwise disjoint and elements of the sample space, then P